1 What is Amazon Web Services?

This chapter covers

- Overview of Amazon Web Services
- The benefits of using Amazon Web Services
- What you can do with Amazon Web Services
- Creating and setting up an AWS account

Almost every IT solution gets labeled with the term *cloud computing* or even just *cloud* nowadays. Buzzwords like these may help sales, but they're hard to work with when trying to teach—or learn—how to work with these technologies. So, for the sake of clarity, let's start this book by defining some terms.

Cloud computing, or the cloud, is a metaphor for supply and consumption of IT resources. The IT resources in the cloud aren't directly visible to the user; layers of abstraction exist in between. The level of abstraction offered by the cloud varies, from offering virtual machines (VMs) to providing Software as a Service (SaaS) based on complex distributed systems. Resources are available on demand in enormous quantities, and you pay for what you use.

The official definition from the National Institute of Standards and Technology follows:

> *Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (networks, virtual machines, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.*